View Reviews

Paper ID
168

Paper Title
DSANet: A Grid-Search Optimized Heterogeneous Ensemble for Ischemic Stroke Lesion Segmentation in Low-Resource Settings

Track Name
MIUA2026

Reviewer #1

Questions

1. Brief overview of the paper.
The paper proposes DSANet, a heterogeneous ensemble framework for ischemic stroke lesion segmentation from multimodal MRI using FLAIR, DWI, and ADC sequences. The method combines three 3D segmentation models: DerNet, SegResNet, and Attention U-Net. Instead of using equal ensemble weights, the authors use a validation-set grid search to optimize both the model fusion weights and the final binarization threshold. The pipeline also uses test-time augmentation, morphological post-processing, mixed-precision training, gradient accumulation, and Dice/Focal loss.

The experiments are conducted on the ISLES 2022 dataset, using 250 public cases split into 70% training, 15% validation, and 15% testing. The authors report a macro Dice score of 0.8780, macro IoU of 0.7826, precision of 0.8333, and recall of 0.9279. They also report subgroup performance, including Dice of 0.7306 for small lesions, 0.8860 for large lesions, 0.9541 for deep brain structures, and 0.8350 for lesion boundaries.

2. What are the main strengths of the paper?
1. Clinically important problem: Ischemic stroke lesion segmentation is highly relevant for diagnosis, treatment planning, and clinical decision support.

2. Multimodal MRI input: Combining FLAIR, DWI, and ADC is appropriate because these modalities provide complementary information for ischemic stroke detection and segmentation.

3. Heterogeneous ensemble design: The use of three different architectures can improve robustness by combining complementary feature extraction mechanisms.

4. Grid-search optimized ensemble: Optimizing ensemble weights and threshold on the validation set is better than simply averaging model outputs.

5. Subgroup analysis included: The paper reports performance across lesion sizes and anatomical regions, which is useful because small lesions and boundary regions are often difficult in stroke segmentation.

6. Qualitative results: The paper includes boundary overlays, error maps, and Layer-CAM visualization, which helps readers inspect the model behavior visually.

3. What are the main weaknesses of the paper?

1. The novelty is limited: The main idea is an ensemble of existing architectures with grid-search weighting.

2. No external validation: The model is evaluated only on an internal 70/15/15 split of ISLES 2022.

3. Some terminology errors: For example, Figure 3 caption says the confusion matrix shows segmentation of brain tumor regions, although the paper is about stroke lesions.

4. Small lesion performance remains weak: The Dice score for small lesions is 0.7306, which is much lower than performance on large lesions.

5. No ablation study: The paper does not clearly isolate the contribution of each component and each architecture.

6. Overall evaluation of the paper.

Weak accept

Reviewer #2

Questions

1. Brief overview of the paper.

The paper proposes DSANet, a heterogeneous 3D ensemble for ischemic stroke lesion segmentation aimed at low computational resource settings. Three branches, a densely connected network ("DerNet"), SegResNet, and Attention U-Net, are trained on multi-modal MRI data stacked into a 3-channel 3D input. Their soft predictions are fused by weighted averaging, with the fusion weights and the binarization threshold selected on the validation set by a constrained grid search (Algorithm 1) under a Dice objective. The pipeline adds test-time augmentation (axis flips) and morphological post-processing.

2. What are the main strengths of the paper?

The paper addresses a relevant and clinically motivated problem practical in low-resource settings where heavy architectures and high-end GPUs are unavailable. The proposed pipeline is assembled from reasonable, well-understood components combined with well known training techniques such as automatic mixed precision and gradient accumulation. The idea of jointly tuning the fusion weights and the binarization threshold on the validation set, rather than relying on uniform or heuristic weighting, has a legitimate kernel, since the two interact. The work also shows good analysis by reporting on fine-grained subgroup analysis across lesion sizes and anatomical regions.

3. What are the main weaknesses of the paper?

Several issues weigh against the paper. First, it is not clear that DSANet is actually better suited to resource-scarce scenarios: the proposed solution is a heterogeneous ensemble of three deep learning models plus a grid-search optimization stage, which can be heavier than the single networks it argues against, and the claimed efficiency is never demonstrated (no compute, memory, or CPU/runtime performance is reported). Second, the novelty is low: by the paper's own related work, ensemble-based stroke segmentation is already established (e.g., DeepISLES on the same dataset, and the MoME mixture-of-experts). Third, the evaluation lacks proper testing and validation against external methods: Tables 3 and 4 report only DSANet's own results, the sole "comparison" (Table 2) is an internal ablation against the model's own components and a single very weak DynUNet baseline which indicates it might not be the correct baseline. No published method is evaluated on the same split — so the claim of outperforming existing approaches is unsupported

6. Overall evaluation of the paper.

Reject

Reviewer #3

Questions

1. Brief overview of the paper.

The paper proposes DSANet, a heterogeneous ensemble of three 3D segmentation networks (DerNet, SegResNet, Attention U-Net) for ischemic stroke lesion segmentation from multi-modal MRI. The ensemble weights and binarisation threshold are jointly optimised via a grid search on the validation set. The method is evaluated on a fixed 70/15/15 split of the ISLES 2022 dataset, reporting a macro Dice of 0.8780. The paper positions itself as a resource-efficient alternative to heavier architectures.

2. What are the main strengths of the paper?

The low-resource framing is a legitimate and underserved problem. Many published stroke segmentation methods do assume access to high-end hardware, and explicitly designing and evaluating under constrained conditions is worth doing.

The grid search ensemble optimisation is at least principled — it avoids arbitrarily assigned weights and the algorithm is described clearly enough to reproduce. The subgroup analysis across lesion sizes and anatomical regions is a useful addition that not all segmentation papers bother with.
The qualitative visualisations in Figures 5 and 6 are reasonable and the boundary analysis panel is informative.

3. What are the main weaknesses of the paper?

The comparability problem is severe and is acknowledged but then essentially dismissed. The results are on a custom 70/15/15 split of ISLES 2022, not the official challenge test set. Every competing method cited in the related work was evaluated on the official leaderboard or a different split. The paper directly compares its Dice of 0.8780 against numbers from Pacal et al. (0.89), Rahman et al. (87.49%), and others in the related work section, but these comparisons are meaningless given the different evaluation protocols. The paper says the custom split "serves as a thorough internal evaluation framework" but then proceeds to frame the results as competitive with the state of the art throughout. This is inconsistent and misleading.
The "low-resource" claim is not backed up by any actual resource measurement. There are no GPU memory figures, no training time comparisons, no FLOPs or parameter counts provided for DSANet versus the methods it claims to be more efficient than. The paper repeatedly asserts it is suitable for low-resource environments but never quantifies what resources it actually requires. This is a central claim of the paper and it is entirely unsubstantiated.
The ensemble weighting result raises an obvious concern that is not addressed. The grid search assigns weight 0.70 to DerNet, 0.10 to SegResNet, and 0.20 to Attention U-Net. DerNet alone achieves a test Dice of 0.8136 (Table 2). The ensemble achieves 0.8780. The question that immediately arises is whether this gain comes from the ensemble principle or simply from the fact that DerNet dominates the ensemble and the other two models provide marginal contributions. The paper does not test DerNet alone with TTA and the optimised threshold, which would be the correct ablation to isolate the ensemble contribution. Without this, the improvement attributed to ensemble fusion may largely be explained by DerNet's individual strength plus TTA and threshold tuning.
The preprocessing description contains a notable inconsistency. The abstract and Section 1 state Z-score normalisation is used, but Figure 1 shows "Min-Max Normalization" in the pipeline diagram, and Figure 4's caption labels it "Z-Score Normalized Input" while showing a range of [-8.0, 526.0] in both panels, which is unusual for Z-score output. It is unclear which normalisation was actually applied.
The writing quality is poor in places. Many sentences are fragmented and repetitive, particularly in Section 3. Phrases like "This study uses", "This step ensures", "This technique ensures" appear dozens of times in quick succession, giving large portions of the paper a mechanical, list-like quality that reads as if it was assembled rather than

written. This affects readability throughout.
The figure caption for Figure 3 refers to "brain tumor regions" rather than stroke lesion regions, which is a clear copy-paste error that suggests insufficient proofreading.
Several references are concerning. Reference 2 is a paper on network intrusion detection, not medical image segmentation — it appears to have been cited for the definition of F1-score, which does not require citation at all, let alone to an unrelated domain. Reference 13 is a time-warping analysis paper cited for Z-score normalisation. These citations suggest the reference list was not carefully curated.
The 64×64×64 input resolution is low for 3D brain MRI and is justified only by GPU memory constraints. There is no analysis of the performance impact of this downsampling relative to full-resolution inputs, nor is there discussion of whether this resolution is sufficient to detect small infarcts reliably — particularly given that small lesion performance (0.7306) is the weakest point of the system.

4. Constructive feedback to authors for improving the paper.

The most important thing the authors should do before resubmission is address the evaluation protocol issue. Either submit predictions to the ISLES 2022 challenge leaderboard for official evaluation, or stop comparing numbers against challenge results. The current situation of claiming competitive performance against methods evaluated on a different test set is not defensible.
The resource efficiency claim needs to be quantified. Report GPU memory usage, training time, and ideally inference time per volume. Compare these against at least one of the methods claimed to be impractical for low-resource settings.
The DerNet-alone-with-TTA ablation should be added to Table 2. Without it, it is impossible to assess how much of the final performance comes from the ensemble versus the dominant constituent.
The normalisation inconsistency needs to be resolved and stated clearly in one place.
The writing should be substantially revised. The fragmented sentence style makes the methodology section particularly difficult to read and gives the impression of padding.
The reference list needs a proper audit. Citing an intrusion detection paper for F1-score and a time-warping paper for Z-score normalisation undermines confidence in the scholarship throughout.

6. Overall evaluation of the paper.

Reject